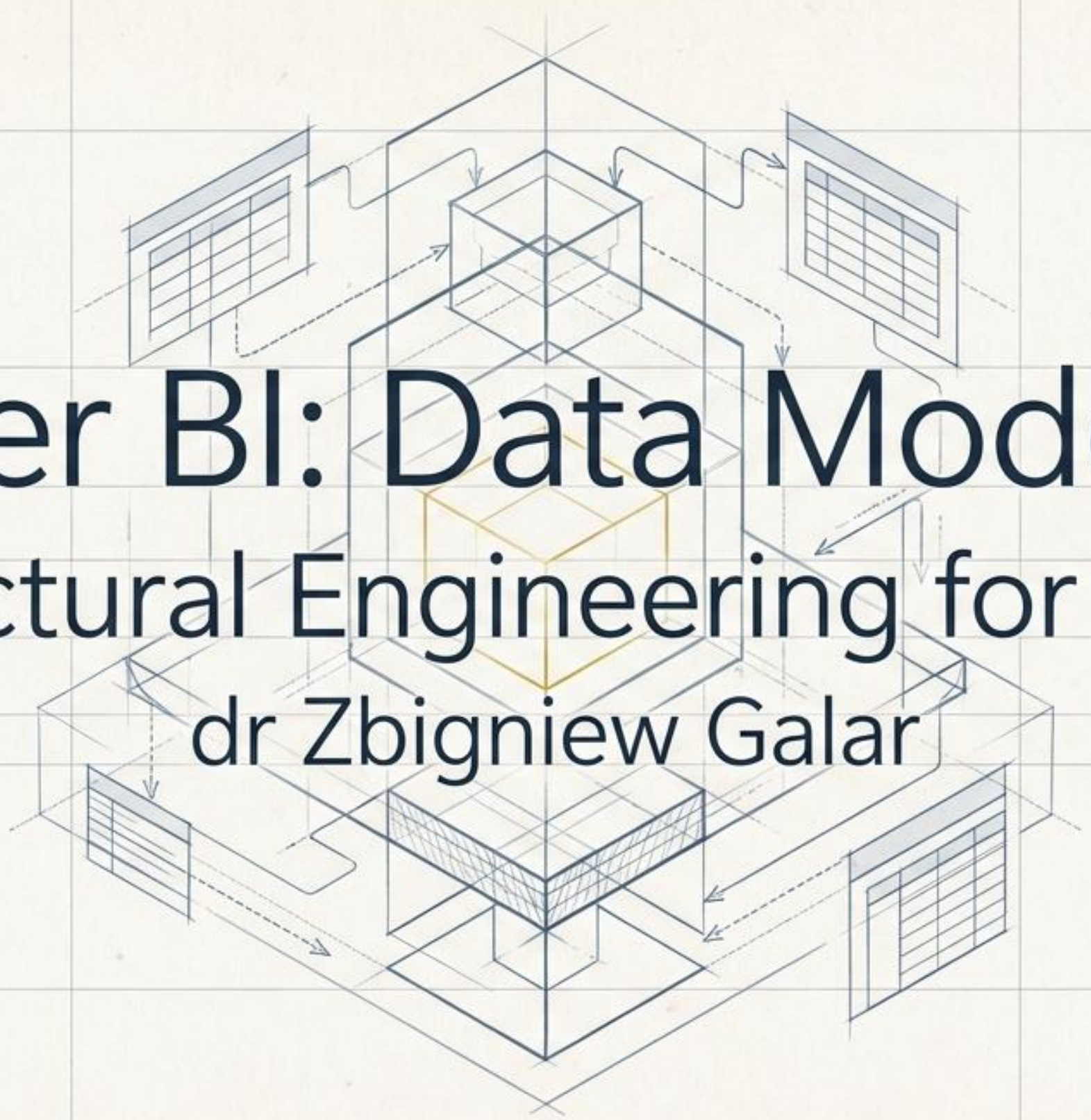




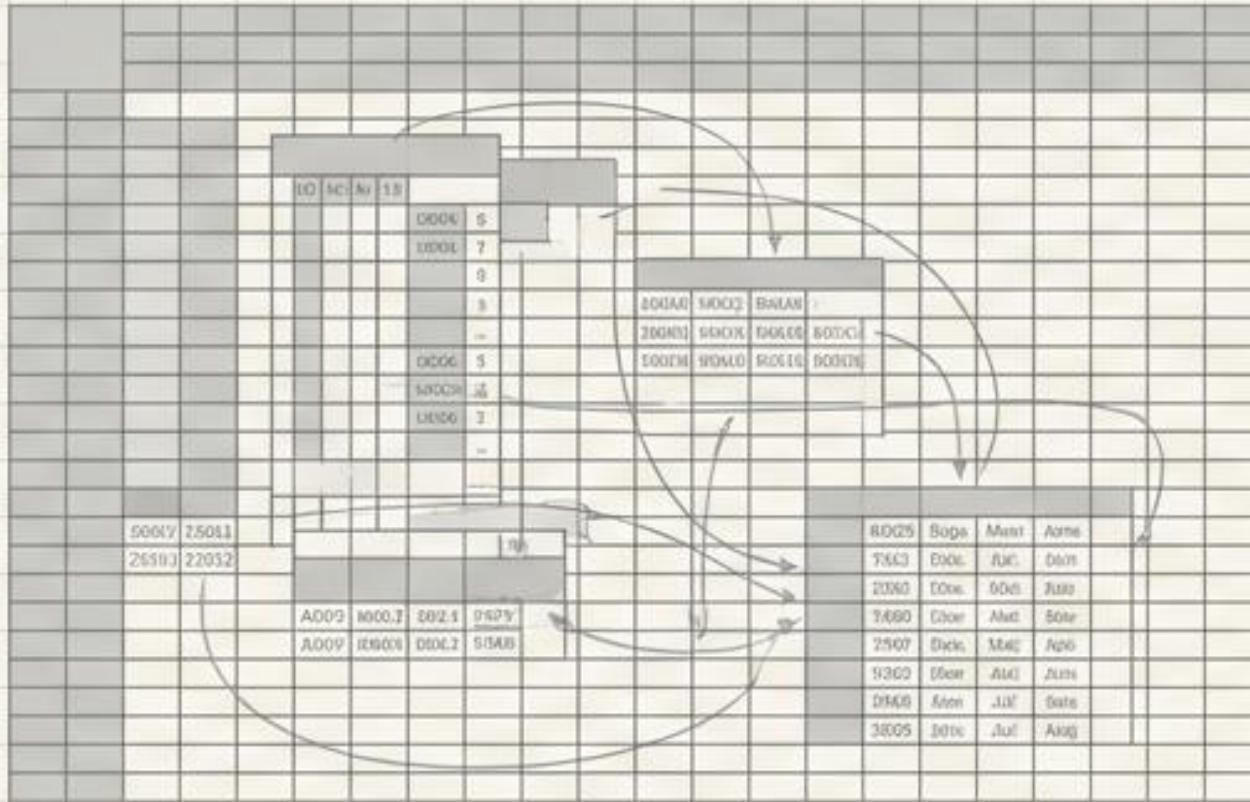
# Power BI: Data Modeling

## Structural Engineering for Data

dr Zbigniew Galar

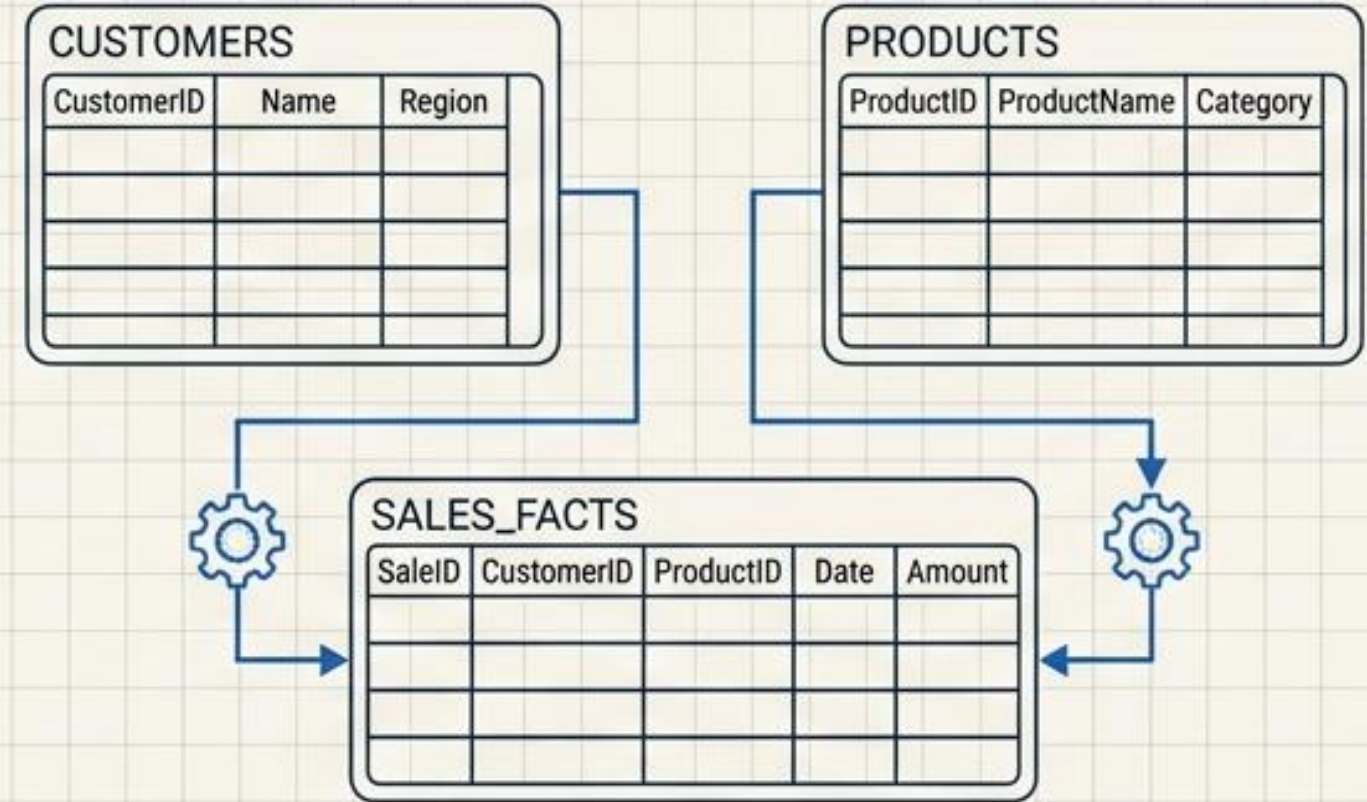


## The Flat World (Excel)



- Requires all data on one sheet
- Relies on heavy LOOKUPS
- Unwieldy and inefficient

## The Relational Architecture (Power BI)

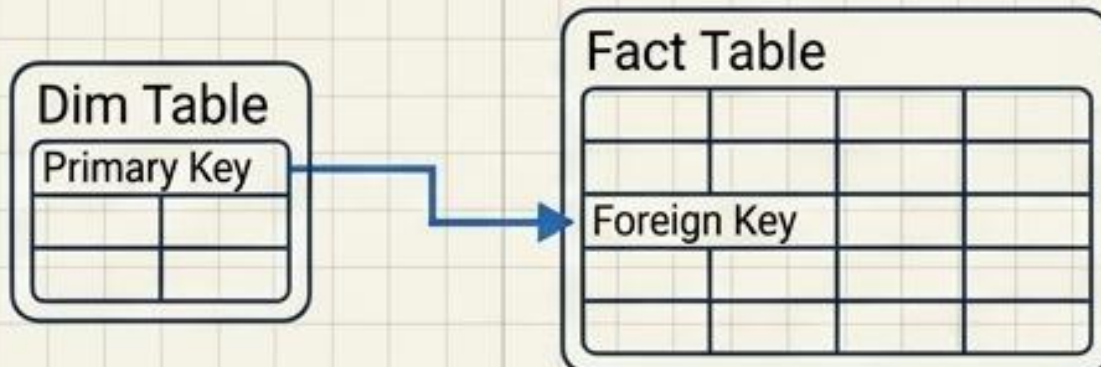


- Logical combinations
- No physical merging
- Highly performant

Takeaway: A good data model is the hidden engine of a performant report.  
A bad model will cripple responsiveness regardless of visual design.



|                         | Dimension Tables (Descriptive)        | Fact Tables (Analytical)                       |
|-------------------------|---------------------------------------|------------------------------------------------|
| Content Type            | Alphanumeric, strings, descriptions   | Numeric values, quantities, prices, dates      |
| Row Count (Cardinality) | One unique row per element            | Many rows per element (recurring events)       |
| Key Type                | Primary Key (Unique IDs)              | Foreign Key (Repeating IDs)                    |
| Visual Role             | Used as labels on charts (axes, rows) | Used as values to be analyzed (averages, sums) |





### Dimension Table (e.g., Product)

| Product Name | ProductKey |
|--------------|------------|
| Widget B     | 123        |
| Widget A     | 123        |

1



\*

### Fact Table (e.g., Sales)

| ProductKey | Date     | Amount |
|------------|----------|--------|
| 123        | 3/1/2021 | 85     |
| 123        | 1/2/2021 | 70     |
| 123        | 3/3/2021 | 123    |
| 123        | 4/4/2021 | 150    |
| 149        | 5/5/2022 | 160    |
| 150        | 6/8/2022 | 200    |
| 166        | 7/2/2022 | 770    |

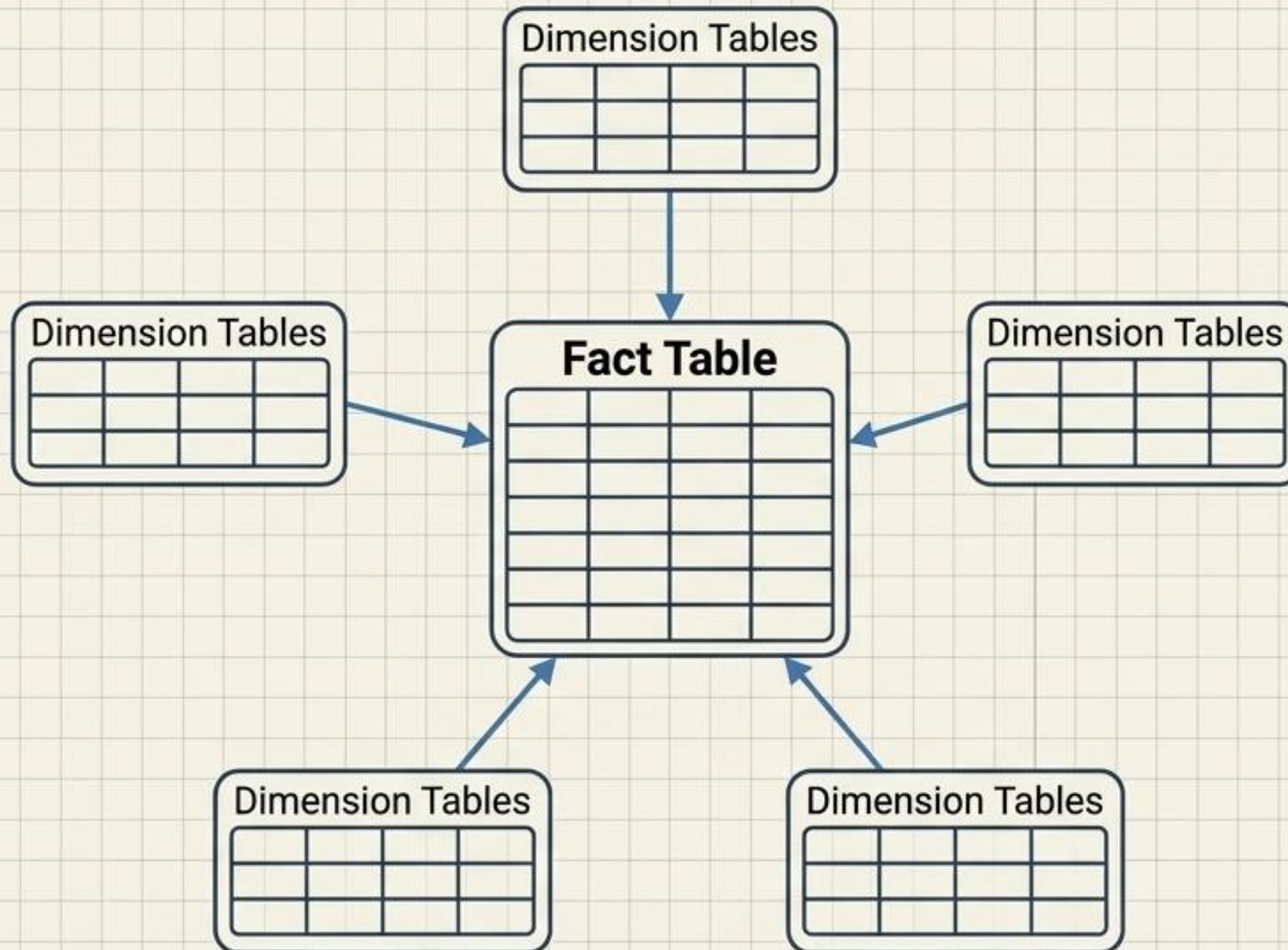
**1. The 1 Side:** A unique selection is made in the Dimension table.

**2. The Transmission:** Relationships are not SQL joins; they are pathways that transmit filters.

**3. The Many Side:** The Fact table is filtered to reveal only the relevant transactions.



# The Star Schema Blueprint



**The Ideal Shape:** The fact table sits at the physical center, surrounded by dimensions. Each dimension table connects directly to the fact table, optimizing the VertiPaq engine for lightning-fast performance.

**Note:** 'Snowflake schemas' (where dimensions connect to other dimensions) exist, but Power BI favors a happy medium. The Star Schema is the ultimate target.



# RELATIONSHIP TAXONOMY

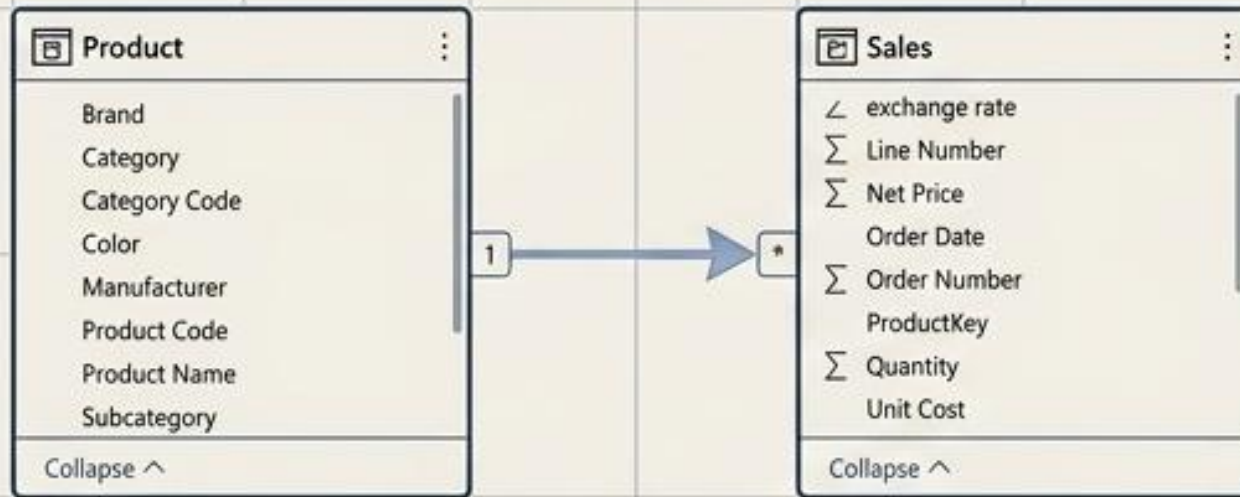
| RELATIONSHIP TYPE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RISK & USE CASE                                                                                                                    | RECOMMENDED ACTION                                                                                            |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|---------------|------------|---------------|------------|---------------|------------|-----------------|------------|-----------------|------------|-----------------|------------|-----------------|--------------|---------------|-----------------|------------|---------------|-----------------|------------|---------------|-----------------|--------------|---------------|-----------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <p>1-to-Many (1-*)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Ideal Use Case: Standard Dim-to-Fact connection.</p> <p>Risk Level: Zero. The gold standard.</p>                                | <p>Use as the primary building block of your model.</p>                                                       |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| <p>Many-to-Many (*-*)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Ideal Use Case: Only when absolutely unavoidable.</p> <p>Risk Level: High. Threat of duplicate counts and slow performance.</p> | <p>Return to Power Query. De-duplicate or add fields to simplify the dimension table into unique records.</p> |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| <p>One-to-One (1-1)</p> <div><table><caption>Dimension Table A</caption><tr><td>Customer A-D</td><td>Attribute 1-4</td></tr><tr><td>Customer B</td><td>Attribute 1-2</td></tr><tr><td>Customer C</td><td>Attribute 1-3</td></tr><tr><td>Customer D</td><td>Attribute 1-4</td></tr></table><table><caption>Dimension Table B</caption><tr><td>Customer A</td><td>Attribute 10-10</td></tr><tr><td>Customer B</td><td>Attribute 10-11</td></tr><tr><td>Customer C</td><td>Attribute 12-12</td></tr><tr><td>Customer D</td><td>Attribute 10-13</td></tr></table></div>  <table><caption>Dimension Table Merged</caption><tr><td>Customer A-D</td><td>Attribute 1-4</td><td>Attribute 10-13</td></tr><tr><td>Customer B</td><td>Attribute 1-4</td><td>Attribute 10-13</td></tr><tr><td>Customer C</td><td>Attribute 1-4</td><td>Attribute 10-13</td></tr><tr><td>Customer A-D</td><td>Attribute 1-4</td><td>Attribute 10-13</td></tr></table> | Customer A-D                                                                                                                       | Attribute 1-4                                                                                                 | Customer B | Attribute 1-2 | Customer C | Attribute 1-3 | Customer D | Attribute 1-4 | Customer A | Attribute 10-10 | Customer B | Attribute 10-11 | Customer C | Attribute 12-12 | Customer D | Attribute 10-13 | Customer A-D | Attribute 1-4 | Attribute 10-13 | Customer B | Attribute 1-4 | Attribute 10-13 | Customer C | Attribute 1-4 | Attribute 10-13 | Customer A-D | Attribute 1-4 | Attribute 10-13 | <p>Ideal Use Case: Two tables with unique rows for the same element.</p> <p>Risk Level: Low, but inefficient.</p> | <p>Streamline the model by merging the two tables in Power Query.</p> |
| Customer A-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Attribute 1-4                                                                                                                      |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 1-2                                                                                                                      |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 1-3                                                                                                                      |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 1-4                                                                                                                      |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 10-10                                                                                                                    |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 10-11                                                                                                                    |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 12-12                                                                                                                    |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 10-13                                                                                                                    |                                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer A-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Attribute 1-4                                                                                                                      | Attribute 10-13                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 1-4                                                                                                                      | Attribute 10-13                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 1-4                                                                                                                      | Attribute 10-13                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |
| Customer A-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Attribute 1-4                                                                                                                      | Attribute 10-13                                                                                               |            |               |            |               |            |               |            |                 |            |                 |            |                 |            |                 |              |               |                 |            |               |                 |            |               |                 |              |               |                 |                                                                                                                   |                                                                       |

**DISCONNECTED TABLES:** If a visual shows endlessly repeating values, check your relationships. A missing connection is almost always the culprit.



# Relationship Directions: Controlling the Flow

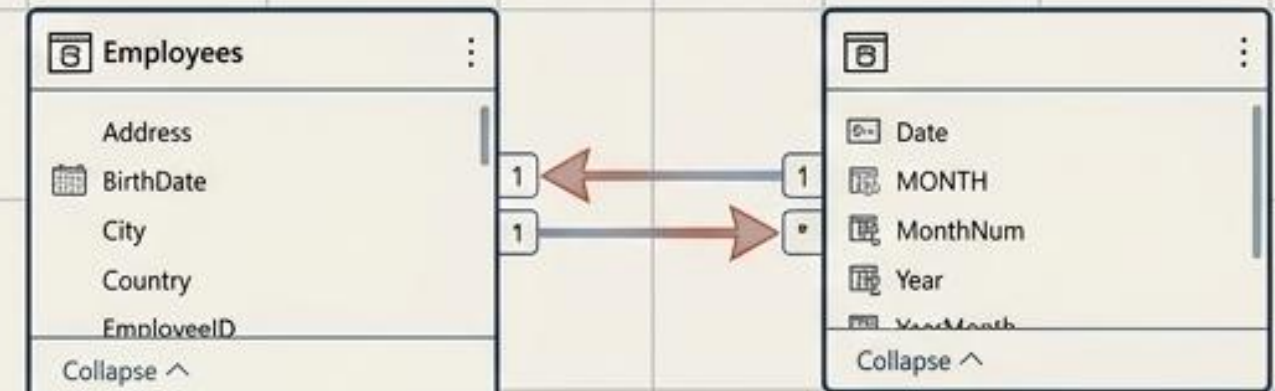
## Single-Direction Relationships (Safe)



Default behavior. Eliminates ambiguity by ensuring filters only flow one way. Highly performant.

The Product dimension table is connected to the Sales fact table with a one-to-many relationship with a single direction filter.

## Bidirectional Relationships (Dangerous)



Tempting, but introduces severe ambiguity, potential calculation errors, and significantly slower report speeds.

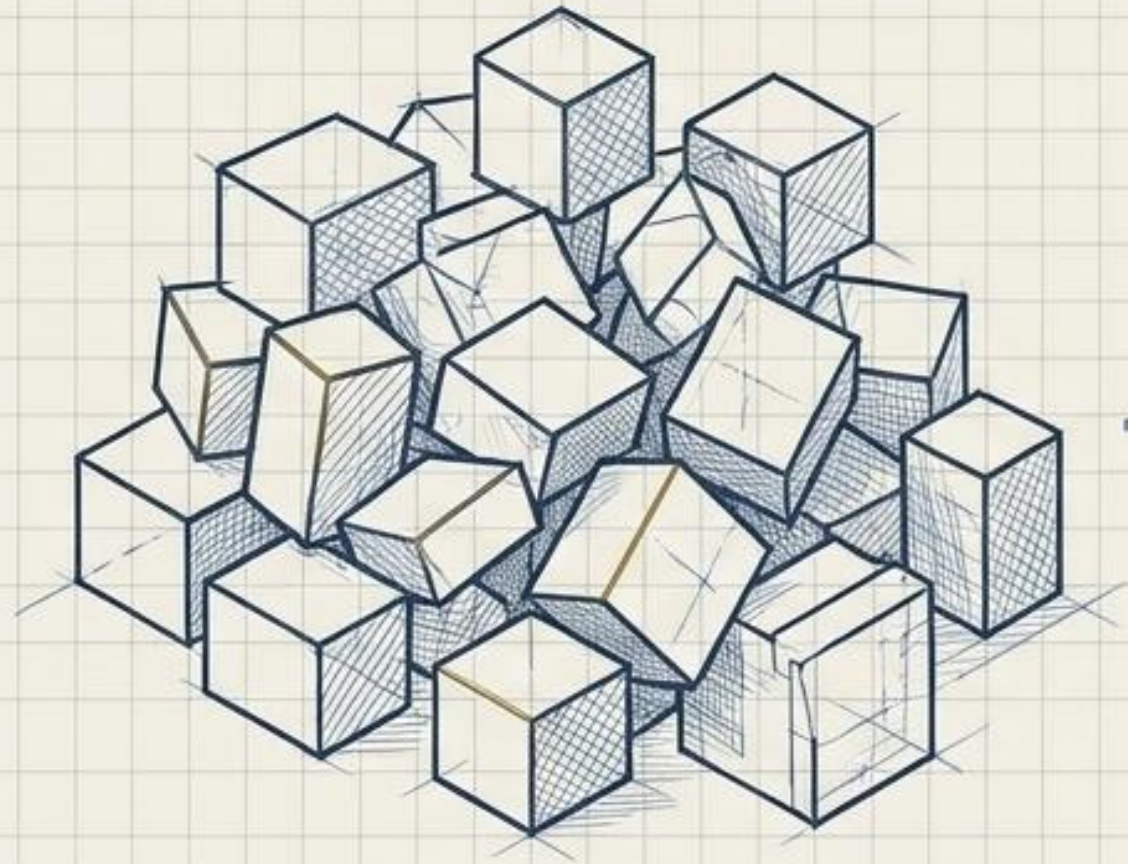
A bi-directional relationship will have arrows pointing in both directions.

**Note:** Many-to-many relationships default to bidirectional. If unavoidable, artificially set the filter direction to mimic a single-direction flow.



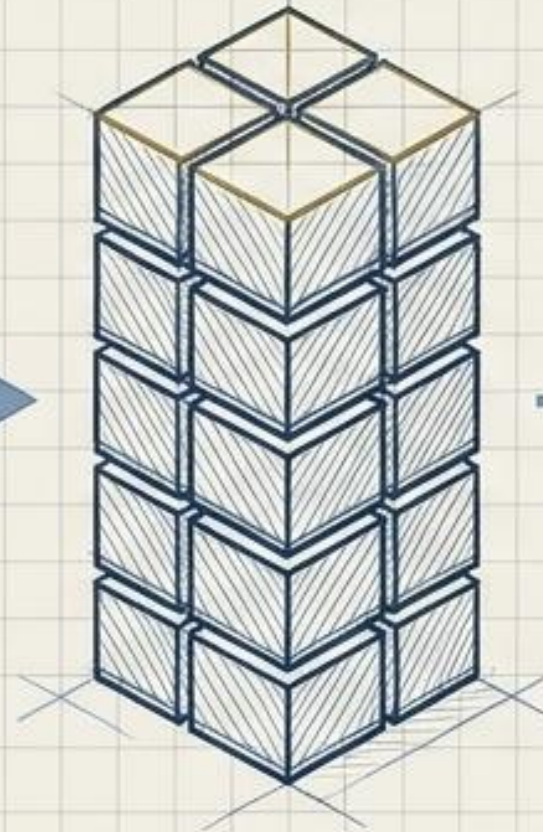
# The VertiPaq Engine & The Cost of Cardinality

**High Cardinality**  
(e.g., Key Columns)



Expensive to store. The engine cannot compress millions of unique values effectively.

**Low Cardinality**  
(e.g., Categories)



Highly compressed. Lightning-fast retrieval and memory efficiency.

## How to Reduce Cardinality

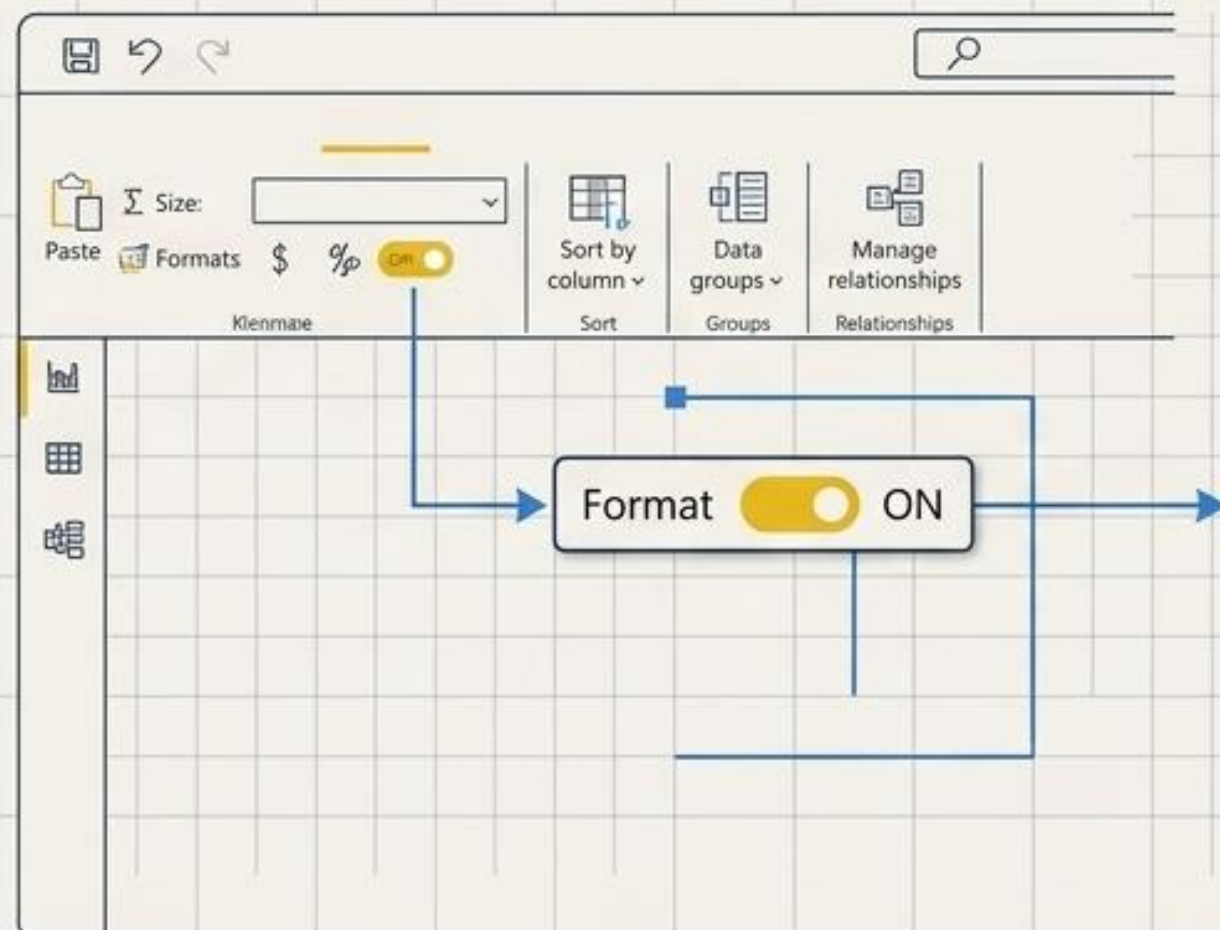
1. Eliminate unnecessary columns entirely.
2. Split complex columns (e.g., separating 'Brand | Category | ID' into three unique columns).
3. Split Date/Time columns into independent Date and Time columns.
4. Convert high-cardinality keys to Integer (Whole Number) data types, which compress far more efficiently than text.



# Desktop Refinements: Formatting vs. Categorization

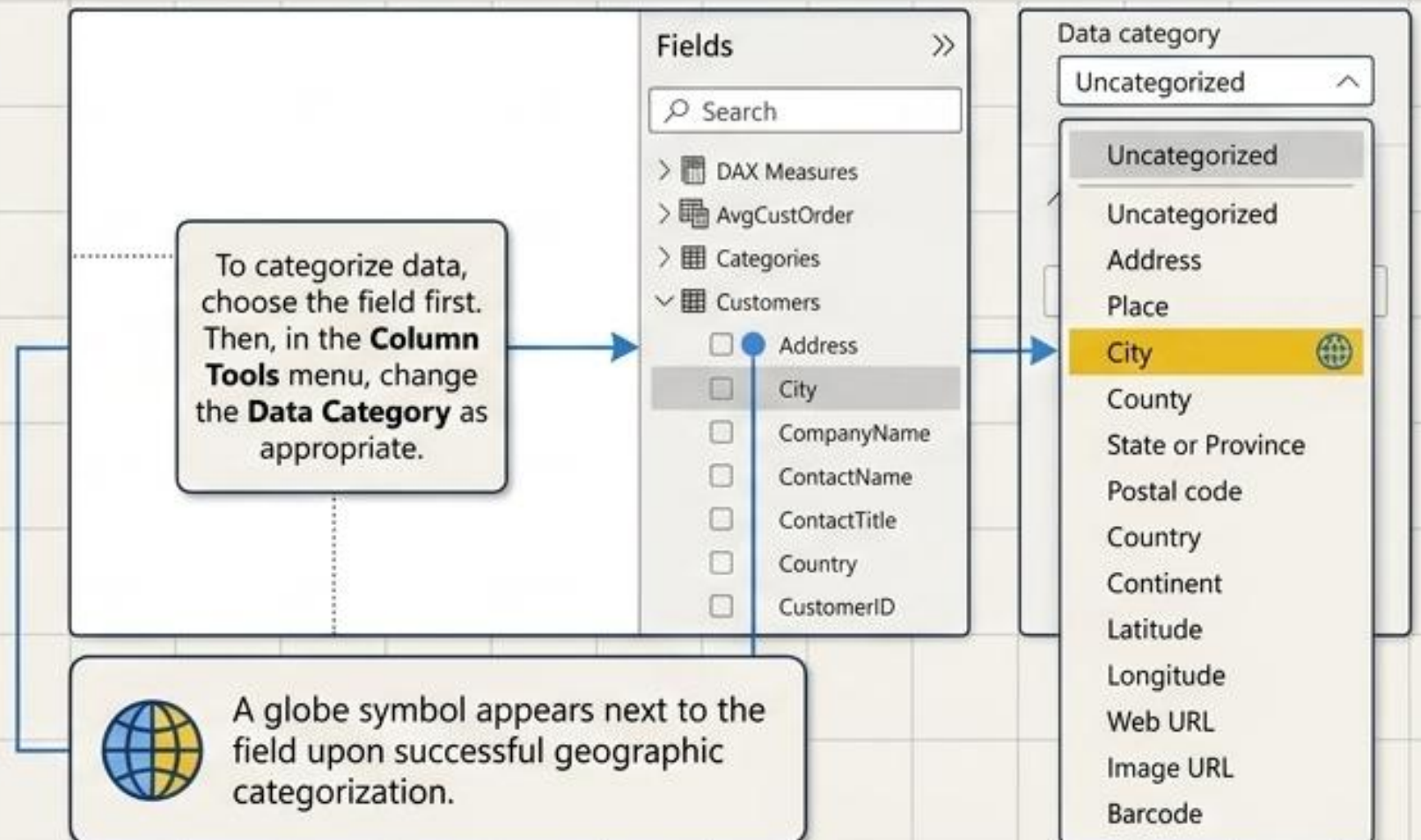
## Formatting

Changes how data displays in the report, not how it is stored.  
(Configured in Desktop, not Power Query).



## Categorization

Gives the engine explicit context to prevent mapping errors (e.g., distinguishing "Columbia" the City from "Columbia" the Country).

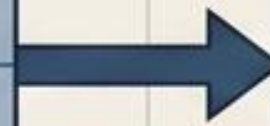




# Installing the Master Clock: The Date Table

## The Problem

Fact tables are chronologically flawed. They skip weekends, miss holidays, ignore custom fiscal years, and completely break DAX Time Intelligence functions.



## The Solution

You must introduce a dedicated, continuous Date Dimension table.

## Implementation Options

### Data Warehouse



Import a pre-built, standardized corporate date table. The ideal, centralized scenario.

### Power Query (M)



Generate via custom M code. Lightweight, efficient, and keeps model size small.

### DAX



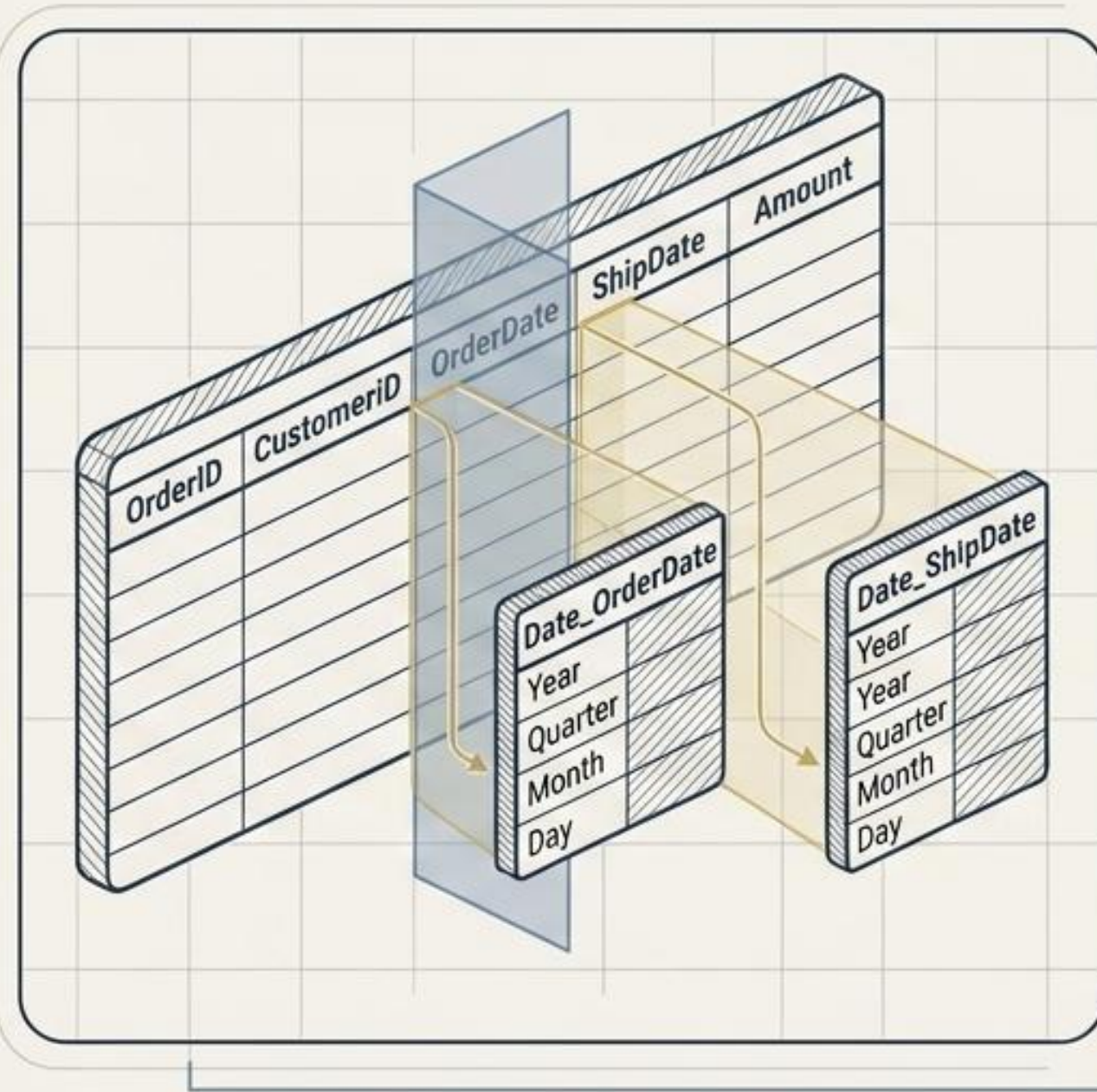
Generate directly in Desktop using DAX functions. Quick to deploy for local files.



# X-Raying the Model: Shadow Table Bloat

## The Threat

If you don't provide a Date Table, Power BI generates a hidden 'shadow' date table for every date field. You cannot edit them, they heavily bloat file size, and they fail reliably with DAX time intelligence.



**The Fix:** Navigate to Options -> Data Load and permanently disable 'Auto date/time' globally to expunge them.



# Finalizing the Date Dimension

## 1. Establish a Sorting Key:

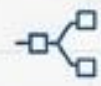
Create a numerical sort column combining Year and Month (e.g, YYYYMM -> 202101) to prevent alphabetical month sorting.



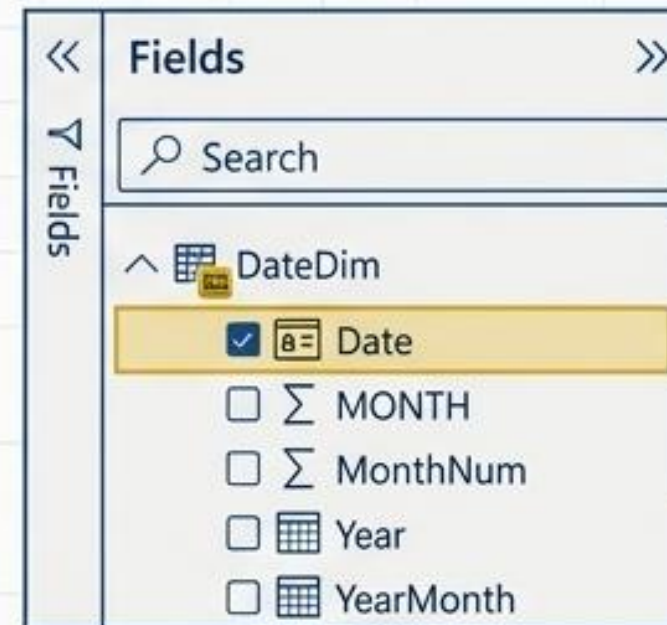
**2. Mark as Date Table:** You must explicitly instruct Power BI to recognize the table. Select the table -> Table Tools -> Mark as Date Table.



**3. Connect:** Link the Date Dim to the Fact table via a standard 1-to-Many relationship.



Select the DateDim table, and in the **Table Tools** menu, choose **Mark As Date Table**. Validate the column that contains one row for every date between the earliest and latest date in the model and has a date data type.



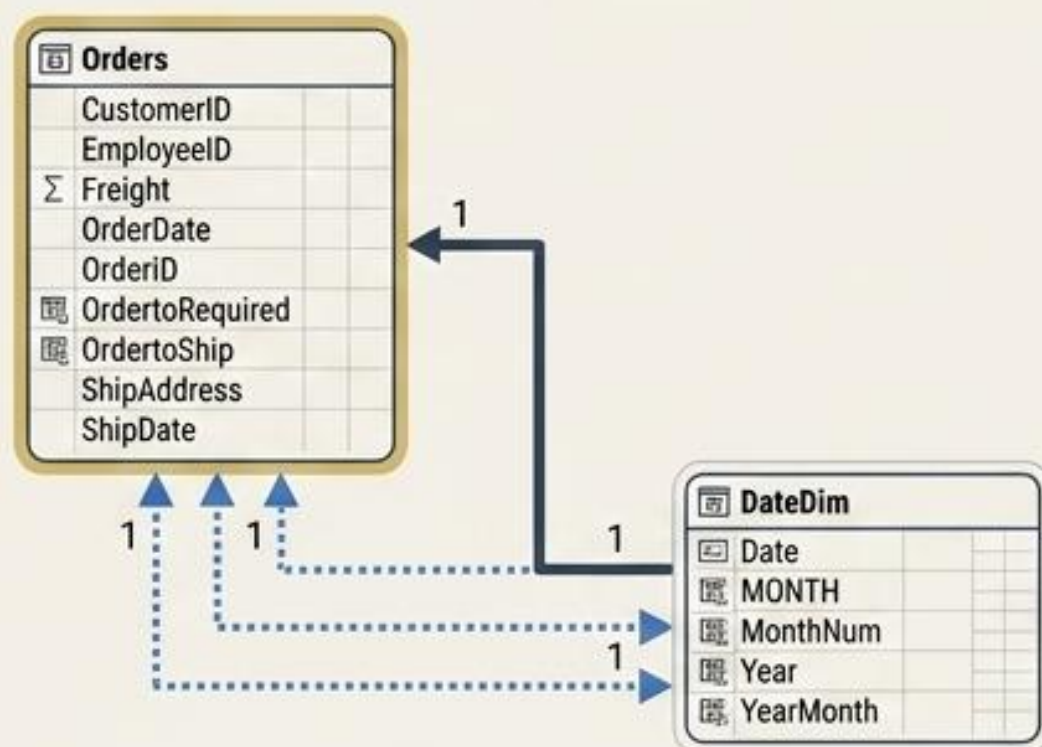
The **Date** column in your date dimension table will have an icon next to it once the date table has been marked as a date table.



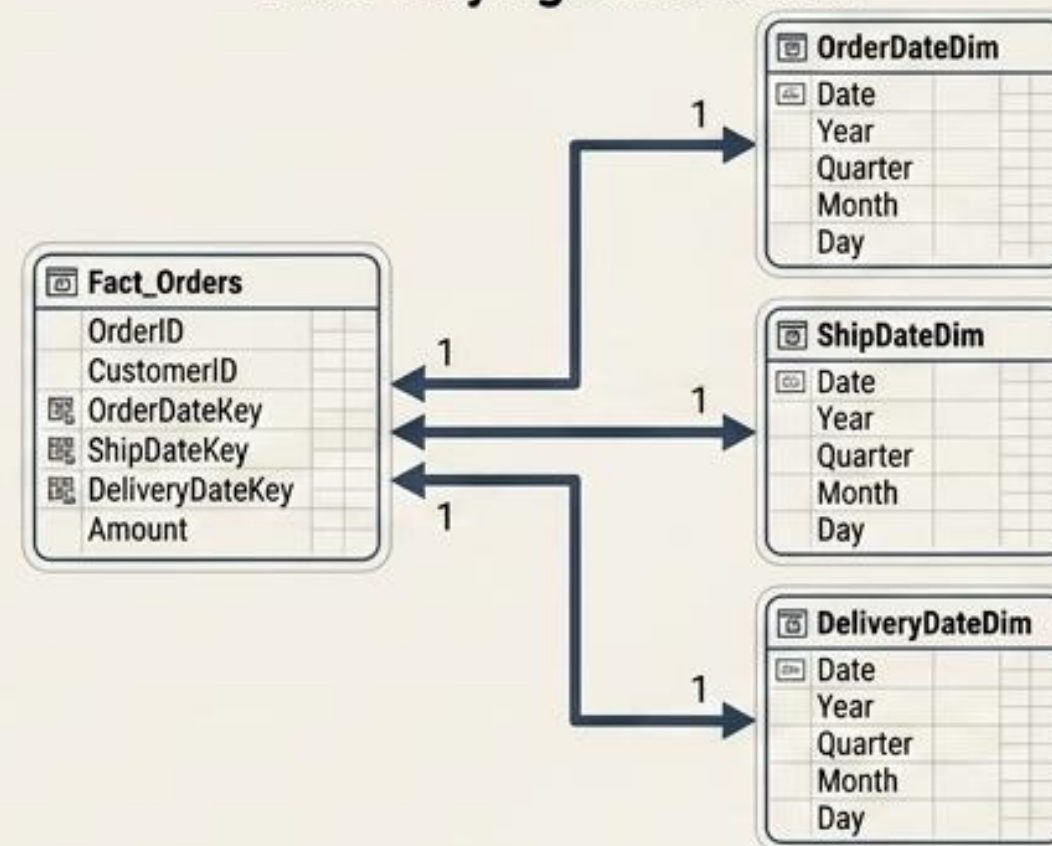
# Handling Multiple Timelines

**Context:** You have Order Date, Ship Date, and Delivery Date. But two tables can only share ONE active relationship.

## Inactive Relationships (Recommended)



## Role-Playing Dimensions



**Mechanism:** Connect the primary date with an Active line. Connect secondary dates with Inactive (dotted) lines. Use the **USERELATIONSHIP** DAX function to temporarily activate them. Highly memory efficient.

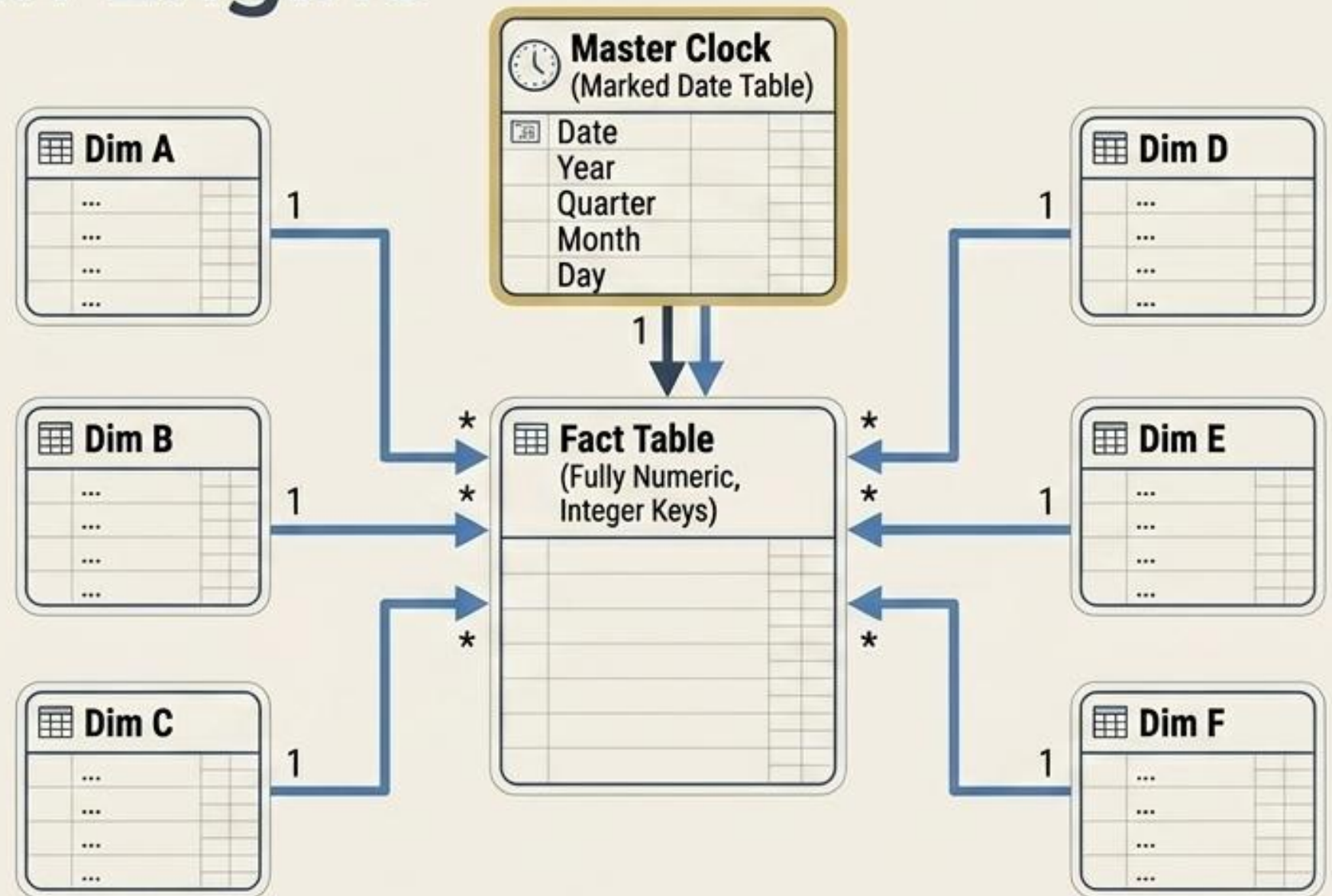
**Mechanism:** Clone the DateDim table entirely. Warning: Consumes significantly more memory. Only viable for very small, simple models.



# Synthesis: The Golden Engine

# The Invisible Rules

- ☒ 1-to-Many Relationships Exclusively
- ☒ Single-Direction Filter Flow
- ☒ Optimized Cardinality (Integers over Strings)
- ☒ Dedicated, Marked Date Table
- ☒ Auto Date/Time Disabled



You will spend **80%** of your time **cleaning and modeling data**.  
Be patient. Build the Golden Engine, and the report will **build itself**.